Aligning a Frozen Transformer with an External Memory

Experiment report — DecisionChains / memory_experiment

1 What we did

A pretrained **12-layer, 512-dim GPT-NeoX** was trained to execute *decision chains*: vectors $v_0 \in \mathbb{Z}_{10}^6$ generate 3–5 blocks, at each step one of two decision functions f, g picks the next letter-op. The two functions are:

$$f(v) = \begin{cases} 10 + v_1 & v_0 \text{ even} \\ v_1 & \text{else} \end{cases} \qquad g(v) = \begin{cases} 10 + v_4 & v_3 \text{ even} \\ v_4 & \text{else} \end{cases}$$

Pretraining uses a fair coin flip between f and g. The resulting model picks each one $\approx 50\%$ of the time per step.

Goal: steer the model to always pick f without modifying the base model.

Memory module. One attention head with trainable keys $K \in \mathbb{R}^{N\times d}$, values $V \in \mathbb{R}^{N\times d}$, query projection $W_q \in \mathbb{R}^{d\times d}$, and optional scalar gate γ:

$$h' \;=\; h \;+\; \gamma \cdot \operatorname{softmax}\!\big(W_q h \cdot K^\top/\sqrt{d}\big)\, V$$

Hooked at layer L; base model frozen. At N=32, d=512: 294K trainable parameters ($\approx 1\%$ of the base model).

Training. Full-sequence CE on 3,000 mixed traces, with labels overridden at decision points to $f(v)$'s letter. AdamW, lr 3×10^{-3}, batch size 4, 10 epochs. $\sim$3 minutes on one A100.

Finetuning baselines. Full FT (37.9M params) and LoRA rank-8 (294K params) in two supervision modes: **A** (ffff-only data, GT labels — standard LM loss) and **B** (all data with f-override at decision points — same signal the memory uses).

2 Evaluation

Two validation sets, both disjoint from training on both letter tuples and input vectors:

- `val_ffff` — 1,998 all-f examples, chain lengths 3–5 uniform.
- `val_full` — 2,000 mixed-pattern examples. The *honest* alignment test: the model encounters hidden states from coin-flip histories the memory may not have trained on.

Metrics (all evaluated via greedy generation):

- **f-selection (%)** — fraction of decision-point *steps* where the model picked f's letter. Per-step average, so longer chains contribute more. Random baseline $\approx 52.5\%$.
- **Full alignment (%)** — fraction of *examples* where *every* step picked f's letter. Strict conjunction: one wrong step scores zero. Baseline $\approx 8.8\%$ on mixed val (expected 0.525^n over chain lengths 3–5).
- **Op. accuracy (%)** — fraction of blocks with correct arithmetic (we re-execute each chosen letter's transformation and compare block text). Must not regress. Baseline $\approx 94.8\%$.

3 Main results

Table 1 compares the five memory variants and four FT baselines at matched compute (n_{train}=3000, epochs=10, batch size 4). Memory runs use 3 seeds; FT runs use 1.

The memory at layers 1 or 2 with N=32 matches or beats every FT variant on alignment while preserving arithmetic $\sim$**4$\times$ better** than the closest LoRA baseline (-1.0 pp vs. -5.5 pp). A single memory entry (N=1, 263K parameters) is the *only* aligning method that *improves* arithmetic (+1.5 pp). Full FT mode B with 37.9M trainable parameters exhibits catastrophic forgetting (-10 pp op. accuracy).

Table 1: Main comparison, full-val (2k mixed-pattern examples). All numbers in %. Subscripts: change vs. pretrained baseline, in percentage points. *ns* = number of seeds. *Full align.* is the strict per-example metric (every step must be f).

Method	**Params**	**Time**	**f-sel (%)**	**Full align (%)**	**Op. acc (%)**
Pretrained (no training)	–	–	52.5	8.8	94.8
Memory, L=2 N=32 (3s)	294K	225s	**97.3** $_{+44.9}$	**88.0** $_{+79.2}$	93.3 $_{-1.5}$
Memory, L=1 N=32 (3s)	294K	177s	97.2 $_{+44.8}$	88.0 $_{+79.1}$	93.8 $_{-1.0}$
Memory, L=0 N=32 (3s)	294K	228s	96.1 $_{+43.6}$	82.5 $_{+73.6}$	91.8 $_{-3.0}$
Memory, L=2 N=1 (3s)	**263K**	208s	94.8 $_{+42.3}$	76.6 $_{+67.7}$	**96.3** $_{+1.5}$
Memory, L=10 N=64 (3s)	327K	114s	65.8 $_{+13.3}$	16.3 $_{+7.5}$	90.5 $_{-4.3}$
FT LoRA, mode B	294K	252s	96.6 $_{+44.1}$	83.5 $_{+74.6}$	89.3 $_{-5.5}$
FT full, mode A	37.9M	272s	96.0 $_{+43.5}$	84.2 $_{+75.3}$	92.6 $_{-2.1}$
FT full, mode B	37.9M	360s	92.9 $_{+40.5}$	69.8 $_{+60.9}$	84.7 $_{-10.0}$
FT LoRA, mode A	294K	246s	81.6 $_{+29.1}$	37.2 $_{+28.4}$	94.3 $_{-0.5}$

ffff val (in-distribution sanity check)

Table 2: ffff val (1998 all-f examples, disjoint from train). Here every aligning method saturates. `complete_solution` (strictest: all ops correct + every letter valid + final vec matches GT output) is the meaningful metric. Baseline ffff: f-sel 76.8, full align 64.8, op. acc 95.1, complete_sol 73.5 (all %).

Method	**f-sel**	**Full align**	**Op. acc**	**Complete sol**
Memory L=2 N=32 (3s)	99.8	99.4	99.8	96.2±0.2
Memory L=1 N=32 (3s)	99.9	99.9	99.8	**96.8**±0.6
Memory L=0 N=32 (3s)	99.9	99.7	99.9	96.8±0.1
Memory L=2 N=1 (3s)	97.5	92.1	97.9	88.9±0.1
Memory L=10 N=64 (3s)	82.3	68.6	94.8	73.1±1.0
FT LoRA, mode B	100.0	100.0	99.9	96.8
FT LoRA, mode A	99.3	98.2	99.8	**97.7**
FT full, mode A	99.9	99.7	99.7	95.8
FT full, mode B	99.7	99.4	99.4	96.0

On ffff val, every aligning method reaches 99%+ alignment (because that is what they were trained for). FT LoRA mode A is actually the single best on `complete_solution`, but it drops to 37% full alignment on mixed val (Table 1) — a gap of 61 pp that reveals the off-path generalisation limit of training on ffff data alone.

4 Ablations

4.1 Layer ablation

Table 3 fixes n_{train}=3000, N=32 and varies only the injection layer.

Layer 1 ≈ Layer 2 > Layer 0 ≫ Layer 10. Injecting into an early residual layer allows the correction to propagate through all downstream blocks. The L=10 configuration (3 seeds) shows high variance (±6.5 pp on f-selection), suggesting the memory loses effectiveness when injected too close to the output.

4.2 Memory size × training size

Table 4 sweeps n_{train} (training examples) and N (memory entries) at L=2. All single-seed, n_{eval}=200.

Table 3: Injection layer. Layers 0–2 evaluated on full 2k val with 3 seeds; layers 4–11 evaluated on earlier Phase-1 sweep (50 examples, 1 seed) — numbers are rougher but trend is clear. Layer 2 is optimal; later layers degrade sharply.

Layer	f-sel Δ (pp)	Full align Δ (pp)	Op. acc Δ (pp)	Eval
0	$+43.6 \pm 0.5$	$+73.6 \pm 1.7$	-3.0 ± 0.9	2k, 3 seeds
1	$+44.8 \pm 0.2$	$+79.1 \pm 1.1$	$\mathbf{-1.0 \pm 0.9}$	2k, 3 seeds
2	$\mathbf{+44.9 \pm 0.3}$	$\mathbf{+79.2 \pm 1.3}$	-1.5 ± 0.1	2k, 3 seeds
4	+38.7	+58.0	−1.1	50, 1 seed
6	+36.7	+52.0	−5.2	50, 1 seed
8	+26.1	+28.0	−1.9	50, 1 seed
10	$+17.4 \pm 0.4$	$+11.3 \pm 5.0$	-2.8 ± 4.8	50, 3 seeds
11	+20.5	+16.0	−4.8	50, 1 seed

Table 4: Full alignment (%) at L=2 across n_{train} (rows) and N (columns). Baseline is 8%. Bold: row best. Shaded cells ≥ 80%.

n_{train}	N=1	N=2	N=4	N=8	N=16	N=32	N=64
100	12.5	12.5	17.0	15.5	**26.0**	24.0	17.5
1000	55.5	60.5	80.0	**80.5**	74.5	79.5	78.0
2000	67.0	86.0	83.5	82.5	**90.5**	89.5	80.5
3000	73.2±2.4	74.0	89.0	**89.5**	87.0	88.0±2.6	89.5

Key pattern. The memory saturates early in N: by N=4–8 we capture most of the gain; by N=16–32 the ceiling is reached. Training data is the harder bottleneck: n=100 plateaus near 26% full alignment; n≥1000 crosses 80%; n=3000 saturates at $\sim$ 91%.

4.3 Ingredient ablation

Table 5 ablates the recipe at L=10, N=32 (Phase-1 sweep, n_{eval}=50). Each row changes one dimension from the reference.

Table 5: What breaks the recipe. Layer 10 reference so differences are bigger. Dropping full-sequence CE destroys arithmetic (−46 pp). Dropping the f-override eliminates the alignment signal.

Config	Change from reference	f-sel Δ	Align Δ	Op Δ
A1_ref	(full recipe, 3 seeds)	$+17.4 \pm 0.4$	$+11.3 \pm 5.0$	-2.8 ± 4.8
A2_no_mix	ffff-only data, GT labels	+3.0	+2.0	+0.1
A3_no_fullseq	DP-only CE	−4.3	−2.0	−46.2
A4_no_gate	no learnable γ	+11.5	+4.0	−1.0
A5_no_override	GT labels, no f-override	−0.3	0.0	+2.2

Order of importance: (1) the f-override at decision points is the actual alignment signal — without it, mixed training does nothing ($A5$, $\Delta \approx 0$); (2) mixing the training distribution matters — ffff-only training generalises poorly to off-path hidden states ($A2$); (3) full-sequence CE is essential — DP-only CE breaks arithmetic catastrophically ($A3$); (4) the gate contributes a small, consistent boost ($A4$).

5 Takeaways

1. **A small attention-head-shaped module, attached at an early residual layer, matches or beats full finetuning on alignment while preserving arithmetic roughly 4× better.** The gain is architectural, not a consequence of matching supervision: LoRA at the same parameter count with the same f-override signal loses by ∼4 pp on op. accuracy.

2. N=1 **is sufficient for high alignment and *improves* arithmetic.** Combined with the linear probe findings earlier in the project, this is consistent with the hypothesis that the f/g preference is controlled by (approximately) a single direction in layer-2 residual space.

3. **Full FT with mode B breaks.** With 37.9M free parameters, the override-on-mixed-data signal causes catastrophic forgetting (-10 pp op. accuracy). LoRA on the same supervision stays stable because the adapter can only move the model in a low-rank subspace.

4. **Early layers dominate.** L$\in \{1, 2\}$ are tied optimal. L$= 10$ (high variance across seeds) barely works. This suggests that for inference-time alignment of this kind, the correction should be injected where the downstream network has most room to propagate it.